

AS Level Mathematics — Pure Mathematics 1 (P1)

Practice Paper 6

Time allowed: **1 hour 50 minutes**

Instructions to candidates

- Answer **all** questions. Write your answers on paper, showing all necessary working. Marks may be lost for insufficient working.
- If a numerical answer is not exact, give it correct to **3 significant figures**; give angles in degrees correct to **1 decimal place**, unless a different level of accuracy is specified in the question.
- A scientific calculator may be used. Do **not** use a graphical or symbolic calculator.
- The number of marks is given in brackets [] at the end of each question or part question. The total mark for this paper is **75**.
- This paper is written in the style of Cambridge International 9709 Paper 1 (Pure Mathematics 1). It is a practice paper, not an official examination paper.

List of formulae (as provided in the examination)

Quadratic equation: for $ax^2 + bx + c = 0$, $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Binomial expansion: $(a + b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \dots + b^n$, where $\binom{n}{r} = \frac{n!}{r!(n-r)!}$

Arithmetic series: $u_n = a + (n-1)d$, $S_n = \frac{1}{2}n(a + l) = \frac{1}{2}n\{2a + (n-1)d\}$

Geometric series: $u_n = ar^{n-1}$, $S_n = \frac{a(1-r^n)}{1-r}$ ($r \neq 1$), $S_\infty = \frac{a}{1-r}$ ($|r| < 1$)

Circular measure: arc length $= r\theta$, sector area $= \frac{1}{2}r^2\theta$ (θ in radians)

Trigonometry: $\tan \theta \equiv \frac{\sin \theta}{\cos \theta}$, $\sin^2 \theta + \cos^2 \theta \equiv 1$

Differentiation and integration: $\frac{d}{dx}(x^n) = nx^{n-1}$, $\int x^n dx = \frac{x^{n+1}}{n+1} + c$ ($n \neq -1$)

1 Find the set of values of x for which $x^2 + 2x - 15 > 0$. **[3]**

2 A sector of a circle has radius 12 cm and arc length 15 cm.

(a) Find the angle of the sector in radians. **[1]**

(b) Find the area of the sector. **[1]**

(c) Find the length of the chord joining the ends of the arc. **[2]**

3 In an arithmetic progression, the 8th term is three times the 2nd term, and the 5th term is 24. Find the sum of the first 20 terms of the progression. **[4]**

4

(a) Find the first three terms, in ascending powers of x , in the expansion of $(2 + x)^6$. [3]

(b) Hence find the coefficient of x^2 in the expansion of $(1 - x)^2(2 + x)^6$. [2]

5 Functions f and g are defined by

$$f(x) = 2x + 5 \quad \text{for } x \in \mathbb{R}, \quad g(x) = \frac{8}{x-1} \quad \text{for } x > 1.$$

(a) Find and simplify an expression for $gf(x)$. [2]

(b) Find an expression for $g^{-1}(x)$ and state the domain of g^{-1} . [3]

(c) State the range of g . [1]

6 The points A and B have coordinates $(-1, 3)$ and $(5, 1)$ respectively.

(a) Find the equation of the perpendicular bisector of AB , giving your answer in the form $y = mx + c$. [3]

(b) The perpendicular bisector meets the y -axis at the point D . Find the area of triangle ABD . [3]

7

(a) Prove the identity $\tan \theta \sin \theta + \cos \theta \equiv \frac{1}{\cos \theta}$. [3]

(b) Hence solve the equation $\tan \theta \sin \theta + \cos \theta = 4 \cos \theta$ for $0^\circ \leq \theta \leq 360^\circ$. [4]

8 A curve has equation $y = \frac{6}{2x-1}$ for $x > \frac{1}{2}$.

(a) Find the equation of the tangent to the curve at the point where $x = 1$. [3]

(b) Explain why the curve is a decreasing function for all $x > \frac{1}{2}$. [1]

(c) Find the equation of the normal to the curve at the point where $x = 2$, giving your answer in the form $ax + by + c = 0$. [3]

9

(a) A curve is such that $\frac{dy}{dx} = 3x^2 - 2x + k$, where k is a constant. The tangent to the curve at the point where $x = 1$ is parallel to the line $y = 5x$. The curve passes through the point $(1, 2)$. Find the value of k and the equation of the curve. [4]

(b) Evaluate $\int_1^4 \left(\sqrt{x} + \frac{1}{\sqrt{x}} \right) dx$, giving your answer as an exact fraction. [4]

10 A ball is dropped from a height of 2 metres. Each time it hits the ground it bounces to a height equal to $\frac{3}{4}$ of the height from which it fell.

(a) Find the height reached after the 3rd bounce. [2]

(b) Find the total distance travelled by the ball from the moment it is dropped until it hits the ground for the 5th time. [3]

(c) Find the total distance travelled by the ball before it comes to rest. [3]

11 The line $y = 2x + c$ and the curve $y = x^2 - x + 3$ are given, where c is a constant.

(a) Find the value of c for which the line is a tangent to the curve. [4]

(b) For this value of c , find the coordinates of the point of contact. [2]

(c) In the case $c = 5$, the line meets the curve at the points P and Q . Find the exact length of PQ . [3]

12 A circle passes through the points $A(2, 1)$ and $B(6, 5)$, and its centre lies on the x -axis.

(a) Find the equation of the circle. [3]

(b) Find the equation of the tangent to the circle at A . [2]

(c) Find the equation of the perpendicular bisector of AB and verify that it passes through the centre of the circle. [3]

END OF PAPER (Total: 75 marks)